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Problem 1

Sets

- $L = \{1, ..., 9\}$: set of **delivery locations**
- $R = \{1, ..., 10\}$: set of **routes**

Indices

- $l \in L$
- $r \in R$

Parameters

- t_r: total duration of route r

$$T = \begin{bmatrix} 6 \\ 4 \\ 7 \\ 5 \\ 4 \\ 6 \\ 5 \\ 3 \\ 7 \\ 6 \end{bmatrix}$$

- $p_{lr} = \begin{cases} 1 \text{ if location } l \text{ is covered by route } r \\ 0 \text{ otherwise} \end{cases}$

$$P = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

Decision Variables

- $x_r = \begin{cases} 1 \text{ if route } r \text{ is selected} \\ 0 \text{ otherwise} \end{cases}$

Model

$$\begin{aligned} \min \quad & \sum_{r \in R} t_r x_r \\ \text{s.t.} \quad & \sum_{r \in R} p_{lr} x_r = 1 \quad \forall l \in L \\ & \sum_{r \in R} x_r = 3 \\ & x_r \in \{0, 1\} \quad \forall r \in R \end{aligned}$$

Solution

Choosing 3 of the 10 candidate routes so that each of the 9 locations is visited exactly once gives a minimum total time of 12 hours.

Truck	Route	Delivery sequence	Time (h)
1	4	B→E→C→I	5
2	5	A→F→H	4
3	8	D→G	3
		Total	**12**

Table 1: Selected routes (one per truck)

Route j	Delivery sequence	Time t_j (h)	x_j
1	A→D→G	6	0
2	F→B→I	4	0
3	H→E→C	7	0
4	B→E→C→I	5	1
5	A→F→H	4	1
6	D→B→E	6	0
7	G→I→C	5	0
8	D→G	3	1
9	A→B→C	7	0
10	H→B→G	6	0

Table 2: All candidate routes and decision variables x_j (selected rows shaded)

Location	A	B	C	D	E	F	G	H	I
Route	5	4	4	8	4	5	8	5	4
Visits	1	1	1	1	1	1	1	1	1

Table 3: Coverage check: each location served exactly once

Problem 2

Sets

- $S = \{0, 1, 2, 3\}$: number of **spots**
- $P = \{1, 2, 3\}$: **products**

Indices

- $s \in S$
- $p \in P$

Parameters

- r_{sp}: profit (in \$M) from giving product p exactly s spots

$$R = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & -1 \\ 3 & 2 & 2 \\ 3 & 3 & 4 \end{bmatrix}$$

Decision Variables

- $x_{sp} = \begin{cases} 1 \text{ if product } p \text{ is shown in } s \text{ spots} \\ 0 \text{ otherwise} \end{cases}$

Model

$$\begin{aligned} \max \quad & \sum_{p \in P} \sum_{s \in S} r_{sp} x_{sp} \\ \text{s.t.} \quad & \sum_{p \in P} \sum_{s \in S} s x_{sp} = 5 \\ & \sum_{s \in S} x_{sp} = 1 \qquad \forall p \in P \\ & x_{sp} \in \{0, 1\} \qquad \forall s \in S, p \in P \end{aligned}$$

Solution

Allocating 5 spots across 3 products (at most 3 each) gives a maximum total profit of \$7 million.

Product	TV spots	Profit (\$M)
Product 1	2	3
Product 2	0	0
Product 3	3	4
Total	**5**	**7**

Table 4: Optimal allocation

Spots s	Product 1	Product 2	Product 3
0	0	0	0
1	1	0	-1
2	3	2	2
3	3	3	4

Table 5: Profit table with chosen levels shaded ($x_{sp} = 1$)

Spots s	x_{s1}	x_{s2}	x_{s3}
0	0	1	0
1	0	0	0
2	1	0	0
3	0	0	1
$\sum_s x_{sp}$	1	1	1

Table 6: Decision variables x_{sp}

Problem 3

Sets

- $W = \{1, 2, 3, 4\}$:
- $R = \{1, 2, 3\}$:

Indices

- $w \in W$:
- $r \in R$:

Parameters

- k_w: capacity of WH w

$$K = \begin{bmatrix} 100 \\ 100 \\ 100 \\ 100 \end{bmatrix}$$

- c_w: fixed cost for WH w

$$C = \begin{bmatrix} 400 \\ 500 \\ 300 \\ 150 \end{bmatrix}$$

t_{wr}: shipping cost from WH w to region r

$$T = \begin{bmatrix} 20 & 40 & 50 \\ 48 & 15 & 26 \\ 26 & 35 & 18 \\ 24 & 50 & 35 \end{bmatrix}$$

- d_r: demand for region r

$$D = \begin{bmatrix} 80 \\ 70 \\ 40 \end{bmatrix}$$

Decision Variables

- x_{wr}: quantity shipped from WH w to region r
- $y_w = \begin{cases} 1 \text{ if WH } w \text{ is opened} \\ 0 \text{ otherwise} \end{cases}$

Model

$$
\begin{aligned}
\min \quad & \sum_{w \in W} \sum_{r \in R} t_{wr} x_{wr} + \sum_{w \in W} c_w y_w \\
\text{s.t.} \quad & \sum_{w \in W} x_{wr} \geq d_r && \forall r \in R \\
& \sum_{r \in R} x_{wr} \leq k_w y_w && \forall w \in W \\
& y_1 \leq y_2 \\
& \sum_{w \in W} y_w \leq 2 \\
& y_2 + y_4 \geq 1 \\
& x_{wr} \geq 0 && \forall w \in W, r \in R \\
& y_w \in \{0, 1\} && \forall w \in W
\end{aligned}
$$

Solution

Opening Los Angeles and Atlanta meets all weekly demand at a minimum cost of $4750 per week ($650 fixed + $4100 shipping).

Warehouse	y_i	Shipped / cap.	Fixed	Shipping	Total
New York	0	0 / 100	—	$0	$0
Los Angeles	1	100 / 100	$500	$1830	$2330
Chicago	0	0 / 100	—	$0	$0
Atlanta	1	90 / 100	$150	$2270	$2420
		Total	**$650**	**$4100**	**$4750**

Table 7: Warehouse decisions y_i and weekly cost by warehouse (open rows shaded)

From \ To	Region 1	Region 2	Region 3	Shipped
New York	0 ($20)	0 ($40)	0 ($50)	0
Los Angeles	0 ($48)	**70** ($15)	**30** ($26)	100
Chicago	0 ($26)	0 ($35)	0 ($18)	0
Atlanta	**80** ($24)	0 ($50)	**10** ($35)	90
Received	80	70	40	
Required	80	70	40	

Table 8: Weekly shipments x_{ij} in units (unit cost in brackets)

Restriction	Constraint	Holds
If New York is opened, Los Angeles must be opened	$y_{\text{NY}} \leq y_{\text{LA}}$	✓
At most two warehouses can be opened	$\sum_i y_i \leq 2$	✓
Either Atlanta or Los Angeles must be opened	$y_{\text{LA}} + y_{\text{ATL}} \geq 1$	✓

Table 9: Logical restrictions

Problem 4

Sets

- $M = \{1, ..., m\}$: manufacturing plants
- $C = \{1, ..., n\}$: customers
- $P = \{1, ..., p\}$: distribution centers

Indices

- $i \in M$
- $k \in C$
- $j \in P$

Parameters

- S_i: annual productive capacity of manufacturing plant i
- D_j: annual demand of customer j
- r: number of distribution centers opened

- c_{ikj}: shipping cost per unit from Plant i to Demand Area j through Distribution Center k
- F_k: fixed operating cost for operating Distribution Center k, if selected
- f_k: variable cost per unit shipped through Distribution Center k

Decision Variables

- x_{ikj}: units shipped from Plant i to Demand Area j through Distribution Center k
- $y_k = \begin{cases} 1 \text{ if Distribution Center } k \text{ is open} \\ 0 \text{ otherwise} \end{cases}$
- $z_{kj} = \begin{cases} 1 \text{ if Distribution Center } k \text{ serves customer } j \\ 0 \text{ otherwise} \end{cases}$

Model

$$
\begin{aligned}
\min \quad & \sum_{i\in I}\sum_{k\in K}\sum_{j\in J}(c_{ikj}+f_k)x_{ikj} + \sum_{k\in K} F_k y_k \\
\text{s.t.} \quad & \sum_{k\in K}\sum_{j\in J} x_{ikj} \le S_i && \forall i \in I && \text{plant capacity} \\
& \sum_{i\in I} x_{ikj} = D_j z_{kj} && \forall k \in K, j \in J && \text{demand met via assigned DC} \\
& \sum_{k\in K} y_k = r && && \text{open exactly } r \text{ DCs} \\
& \sum_{k\in K} z_{kj} = 1 && \forall j \in J && \text{single sourcing} \\
& z_{kj} \le y_k && \forall k \in K, j \in J && \text{assign only to open DCs} \\
& x_{ikj} \ge 0 && \forall i, k, j \\
& y_k \in \{0,1\} && \forall k \in K \\
& z_{kj} \in \{0,1\} && \forall k \in K, j \in J
\end{aligned}
$$

Problem 5

Sets

- $I = \{A, B, M, P, S\}$: cities (demand and possible Domes locations)
- $K = \{40, 20\}$: container types

Indices

- $i, j \in I$: cities
- $k \in K$: container type

Parameters

- d_i: monthly demand in city i

$$
d = \begin{bmatrix} 420 \\ 870 \\ 1250 \\ 930 \\ 1310 \end{bmatrix}
$$

- p_i: Domes unit production cost in city i

$$
p = \begin{bmatrix} 450 \\ 480 \\ 505 \\ 490 \\ 515 \end{bmatrix}
$$

- $p^0 = 440$: Ovis unit purchase price (DDP)
- f_{ik}: sea freight per type-k container from Ovis to city i (columns: 40′, 20′)

$$
f = \begin{bmatrix} 2000 & 1200 \\ 1600 & 1000 \\ 1800 & 1100 \\ 1200 & 700 \\ 1650 & 1050 \end{bmatrix}
$$

- g_i: LCL sea freight per unit from Ovis to city i

$$g = \begin{bmatrix} 25 \\ 20 \\ 23 \\ 15 \\ 22 \end{bmatrix}$$

- q_k: capacity of a type-k container (units)

$$q = \begin{bmatrix} 200 \\ 100 \end{bmatrix}$$

- τ_i: sea lead time from Ovis to city i (days)

$$\tau = \begin{bmatrix} 30 \\ 21 \\ 28 \\ 18 \\ 25 \end{bmatrix}$$

- c_{ij}: road transport cost per unit from city i to city j

$$c = \begin{bmatrix} 0 & 35 & 10 & 35 & 25 \\ 35 & 0 & 25 & 70 & 15 \\ 10 & 25 & 0 & 45 & 15 \\ 35 & 70 & 45 & 0 & 55 \\ 25 & 15 & 15 & 55 & 0 \end{bmatrix}$$

- $h = 0.15$: annual interest rate for inventory holding
- $U = 500$: monthly Domes production (fixed commitment)

Decision Variables

- $x_{ik} \in \mathbb{Z}_{\geq 0}$: type-$k$ containers shipped from Ovis to city i
- $y_i \in \mathbb{Z}_{\geq 0}$: units shipped by LCL from Ovis to city i
- $z_{ij} \in \mathbb{Z}_{\geq 0}$: units moved by road from city i to city j
- $w_i \in \{0, 1\}$: 1 if Domes produces in city i, 0 otherwise
- $v_i \geq 0$: units Domes produces in city i

Models

1. Current practice: ship directly from Ovis (Q1)

$$\begin{aligned}
\min \quad & \sum_{i \in I} \sum_{k \in K} q_k x_{ik} p^0 + && \textbf{purchase cost (container)} \\
& \sum_{i \in I} \sum_{k \in K} f_{ik} x_{ik} + && \textbf{freight cost (container)} \\
& \sum_{i \in I} p^0 y_i + && \textbf{purchase cost (LCL)} \\
& \sum_{i \in I} g_i y_i + && \textbf{freight cost (LCL)} \\
& \sum_{i \in I} \sum_{k \in K} \frac{h}{365} p^0 \tau_i x_{ik} q_k + && \textbf{holding cost (container)} \\
& \sum_{i \in I} \frac{h}{365} p^0 \tau_i y_i && \textbf{holding cost (LCL)} \\
\text{s.t.} \quad & \sum_{k \in K} x_{ik} q_k + y_i \geq d_i \qquad \forall i \in I && \text{demand} \\
& x_{ik} \in \mathbb{Z}_{\geq 0} \qquad \forall i \in I, k \in K \\
& y_i \in \mathbb{Z}_{\geq 0} \qquad \forall i \in I
\end{aligned}$$

2. Ovis with road rebalancing, no Domes (Q2)

$$
\begin{aligned}
\min \quad & \sum_{i\in I}\sum_{k\in K} q_k x_{ik} p^0 + && && \textbf{purchase cost (container)} \\
& \sum_{i\in I}\sum_{k\in K} f_{ik} x_{ik} + && && \textbf{freight cost (container)} \\
& \sum_{i\in I} p^0 y_i + && && \textbf{purchase cost (LCL)} \\
& \sum_{i\in I} g_i y_i + && && \textbf{freight cost (LCL)} \\
& \sum_{i\in I}\sum_{k\in K} \frac{h}{365} p^0 \tau_i x_{ik} q_k + && && \textbf{holding cost (container)} \\
& \sum_{i\in I} \frac{h}{365} p^0 \tau_i y_i + && && \textbf{holding cost (LCL)} \\
& \sum_{i\in I}\sum_{j\in I} c_{ij} z_{ij} && && \text{transportation cost (road)} \\
\text{s.t.} \quad & \sum_{k\in K} q_k x_{ik} + y_i + \sum_{j\in I} z_{ji} \geq d_i + \sum_{j\in I} z_{ij} && \forall i \in I && \text{flow balance} \\
& x_{ik} \in \mathbb{Z}_{\geq 0} && \forall i \in I, k \in K \\
& y_i \in \mathbb{Z}_{\geq 0} && \forall i \in I \\
& z_{ij} \in \mathbb{Z}_{\geq 0} && \forall i, j \in I
\end{aligned}
$$

3. Ovis with road rebalancing and Domes (Q3, Q4)

$$
\begin{aligned}
\min \quad & \sum_{i\in I}\sum_{k\in K} q_k x_{ik} p^0 + && && \textbf{purchase cost (container)} \\
& \sum_{i\in I}\sum_{k\in K} f_{ik} x_{ik} + && && \textbf{freight cost (container)} \\
& \sum_{i\in I} p^0 y_i + && && \textbf{purchase cost (LCL)} \\
& \sum_{i\in I} g_i y_i + && && \textbf{freight cost (LCL)} \\
& \sum_{i\in I}\sum_{k\in K} \frac{h}{365} p^0 \tau_i x_{ik} q_k + && && \textbf{holding cost (container)} \\
& \sum_{i\in I} \frac{h}{365} p^0 \tau_i y_i + && && \textbf{holding cost (LCL)} \\
& \sum_{i\in I}\sum_{j\in I} c_{ij} z_{ij} + && && \text{transportation cost (road)} \\
& \sum_{i\in I} p_i v_i && && \text{production cost} \\
\text{s.t.} \quad & \sum_{k\in K} q_k x_{ik} + y_i + v_i + \sum_{j\in I} z_{ji} \geq d_i + \sum_{j\in I} z_{ij} && \forall i \in I && \text{flow balance} \\
& \sum_{i\in I} w_i = 1 && && \text{only one Domes site} \\
& v_i = U w_i && \forall i \in I && \text{produce exactly } U \text{ at the open site} \\
& x_{ik} \in \mathbb{Z}_{\geq 0} && \forall i \in I, k \in K \\
& y_i \in \mathbb{Z}_{\geq 0} && \forall i \in I \\
& z_{ij} \in \mathbb{Z}_{\geq 0} && \forall i, j \in I \\
& w_i \in \{0, 1\} && \forall i \in I \\
& v_i \geq 0 && \forall i \in I
\end{aligned}
$$

Cost comparison (monthly, $)

Cost	(1) Direct	(2) + Road	(3) + Domes	(3) – (2)
purchase	2,103,200	2,103,200	1,905,200	-198,000
freight	41,370	40,290	35,100	-5,190
holding	20,860	20,827	18,397	-2,430
road	0	650	950	300
domes	0	0	202,500	202,500
Total	**2,165,430**	**2,164,967**	**2,162,147**	**-2,820**

Table 10: Summary

1. Shipping everything directly from Ovis to each city costs Oz Sourcing 2,165,430 per month.
2. No. Without Domes, it is cheaper to ship a 20′ container to Brisbane instead of LCL and move the surplus by road. This costs 2,164,967 per month, which saves 463 compared with the current practice.
3. Oz Sourcing should ask Domes to manufacture in **Adelaide** (450 units per month).
4. Engaging Domes saves 3,283 per month compared with the current practice. Of this, 463 comes from road rebalancing alone, so the saving attributable to Domes is 2,820 per month.

City	Demand	40′	20′	LCL	Domes	In	Out	$/unit
Adelaide	420	2	0	20	0	0	0	456.14
Brisbane	870	4	0	70	0	0	0	452.76
Melbourne	1250	6	0	50	0	0	0	454.62
Perth	930	4	1	30	0	0	0	449.65
Sydney	1310	6	1	10	0	0	0	453.05

Table 11: Per-city plan, (1) direct from Ovis

City	Demand	40′	20′	LCL	Domes	In	Out	$/unit
Adelaide	420	2	0	20	0	0	0	456.14
Brisbane	870	4	1	0	0	0	30	468.35
Melbourne	1250	6	0	30	0	20	0	447.13
Perth	930	4	1	30	0	0	0	449.65
Sydney	1310	6	1	0	0	10	0	449.49

Table 12: Per-city plan, (2) Ovis + road

City	Demand	40′	20′	LCL	Domes	In	Out	$/unit
Adelaide	420	0	0	0	450	0	30	482.86
Brisbane	870	4	1	0	0	0	30	468.35
Melbourne	1250	6	0	0	0	50	0	435.90
Perth	930	4	1	30	0	0	0	449.65
Sydney	1310	6	1	0	0	10	0	449.49

Table 13: Per-city plan, (3) Ovis + road + Domes

From	To	Units
Brisbane	Melbourne	20
Brisbane	Sydney	10

Table 14: Road transfers, (2) Ovis + road

From	To	Units
Adelaide	Melbourne	30
Brisbane	Melbourne	20
Brisbane	Sydney	10

Table 15: Road transfers, (3) Ovis + road + Domes

Appendix

Problem 1

```
import numpy as np
import gurobipy as gp

P = np.array([
    [1, 0, 0, 0, 1, 0, 0, 0, 1, 0],
    [0, 1, 0, 1, 0, 1, 0, 0, 1, 1],
    [0, 0, 1, 1, 0, 0, 1, 0, 1, 0],
```

```
    [1, 0, 0, 0, 0, 1, 0, 1, 0, 0],
    [0, 0, 1, 1, 0, 1, 0, 0, 0, 0],
    [0, 1, 0, 0, 1, 0, 0, 0, 0, 0],
    [1, 0, 0, 0, 0, 0, 1, 1, 0, 1],
    [0, 0, 1, 0, 1, 0, 0, 0, 0, 1],
    [0, 1, 0, 1, 0, 0, 1, 0, 0, 0],
])

T = np.array([
    6,
    4,
    7,
    5,
    4,
    6,
    5,
    3,
    7,
    6,
])

MAX = 3

m = gp.Model("speedy")

x = m.addMVar(T.shape, vtype=gp.GRB.BINARY, name="x")

m.setObjective((T * x).sum())

m.addConstr(P @ x == 1, name="")
m.addConstr(x.sum() == MAX, name="")

m.optimize()
```

Problem 1

Route j	1	2	3	4	5	6	7	8	9	10
Time t_j (hours)	6	4	7	5	4	6	5	3	7	6
Decision variable x_j (1 = route selected)	0	0	0	1	1	0	0	1	0	0

Delivery location i (a_ij = 1 if route j visits i)	1	2	3	4	5	6	7	8	9	10	LHS		RHS
A	1				1				1		1	=	1
B		1		1		1			1	1	1	=	1
C			1	1			1		1		1	=	1
D	1					1		1			1	=	1
E			1	1		1					1	=	1
F		1			1						1	=	1
G	1						1	1		1	1	=	1
H			1		1					1	1	=	1
I		1		1			1				1	=	1
Number of routes selected (drivers available)											3	=	3

Total time Z (hours) - minimize	12	



Problem 2

```
import numpy as np
import gurobipy as gp

R = np.array([
    [0, 0, 0],
    [1, 0, -1],
    [3, 2, 2],
    [3, 3, 4],
])

S, P = R.shape

S = np.arange(S)

m = gp.Model("supersuds")

x = m.addMVar(R.shape, vtype=gp.GRB.BINARY, name="x")

m.setObjective((R * x).sum(), sense=gp.GRB.MAXIMIZE)
```

```
20
21 m.addConstr((S @ x).sum() == 5)
22 m.addConstr(x.sum(axis=0) == 1)
23
24 m.optimize()
25
```

Problem 2

Profit (millions)

Number of TV spots	Product 1	Product 2	Product 3
0	0	0	0
1	1	0	-1
2	3	2	2
3	3	3	4

Decision variables y_ij = 1 if product i receives j spots

Number of TV spots	Product 1	Product 2	Product 3
0	**0**	**1**	**0**
1	**0**	**0**	**0**
2	**1**	**0**	**0**
3	**0**	**0**	**1**
Spots allocated	2	0	3
Profit by product (millions)	3	0	4

Constraints

Constraint	LHS		RHS
Product 1: exactly one option chosen	1	=	1
Product 2: exactly one option chosen	1	=	1
Product 3: exactly one option chosen	1	=	1
Total TV spots purchased	5	=	5

Total profit Z (millions) - maximize	**7**



Problem 3

```
1  import numpy as np
2  import gurobipy as gp
3
4  K = np.array([
5      100,
6      100,
7      100,
8      100,
9  ])
10
11 C = np.array([
12     400,
13     500,
14     300,
15     150,
16 ])
17
18 T = np.array([
19     [20, 40, 50],
20     [48, 15, 26],
21     [26, 35, 18],
22     [24, 50, 35],
23 ])
24
25 D = np.array([
26     80,
27     70,
28     40,
29 ])
30
31 m = gp.Model("wh")
32
33 x = m.addMVar(T.shape, name="x")
34 y = m.addMVar(K.shape, vtype=gp.GRB.BINARY, name="y")
35
36 m.setObjective((T * x).sum() + (C * y).sum(), sense=gp.GRB.MINIMIZE)
37
38 m.addConstr(x.sum(axis=0) >= D, name="demand")
39 m.addConstr(x.sum(axis=1) <= K * y, name="capacity")
40 m.addConstr(y[0] <= y[1], name="ny_implies_la")
41 m.addConstr(y.sum() <= 2, name="max_two")
42 m.addConstr(y[1] + y[3] >= 1, name="atl_or_la")
```

```
43
44 m.optimize()
```

Problem 3

Data

From \ To (cost per unit, $)	Region 1	Region 2	Region 3	Capacity (units/week)	Fixed cost ($/week)
New York	20	40	50	100	400
Los Angeles	48	15	26	100	500
Chicago	26	35	18	100	300
Atlanta	24	50	35	100	150
Demand (units/week)	80	70	40		



Decision variables

Warehouse	Region 1	Region 2	Region 3	Open? (y_k)	Units shipped		Capacity x open
New York	**0**	**0**	**0**	**0**	0	<=	0
Los Angeles	**0**	**70**	**30**	**1**	100	<=	100
Chicago	**0**	**0**	**1.4211E-14**	**0**	1.42109E-14	<=	0
Atlanta	**80**	**0**	**10**	**1**	90	<=	100
Units received (LHS)	80	70	40				
	=	=	=				
Demand (RHS)	80	70	40				

Logical constraints

Constraint	LHS		RHS
1) If New York is open, Los Angeles is open	0	<=	1
2) At most two warehouses open	2	<=	2
3) Atlanta or Los Angeles open	2	>=	1

Fixed cost ($/week)	650
Shipping cost ($/week)	4,100
Total weekly cost Z ($) - minimize	**4,750**

Problem 5

```
1  from pathlib import Path
2
3  import gurobipy as gp
4  import numpy as np
5  import tomli_w
6
7  # ------------------------------------------------------------------------
8  # Data
9  # ------------------------------------------------------------------------
10 cities = ["Adelaide", "Brisbane", "Melbourne", "Perth", "Sydney"]
11
12 d   = np.array([420, 870, 1250, 930, 1310])        # monthly demand
13 p   = np.array([450, 480, 505, 490, 515])         # Domes unit cost by site
14 p0  = 440                                         # Ovis delivered unit cost
15 f   = np.array([                                  # $/container [40', 20']
16     [2000, 1200],
17     [1600, 1000],
18     [1800, 1100],
19     [1200,  700],
20     [1650, 1050],
21 ])
22 g   = np.array([25, 20, 23, 15, 22])              # $/LCL unit
23 q   = np.array([200, 100])                        # container capacity
24 tau = np.array([30, 21, 28, 18, 25])             # sea lead time (days)
25 c   = np.array([                                  # road $/unit, origin x dest
26     [ 0, 35, 10, 35, 25],
27     [35,  0, 25, 70, 15],
28     [10, 25,  0, 45, 15],
29     [35, 70, 45,  0, 55],
30     [25, 15, 15, 55,  0],
31 ])
32 h = 0.15                                           # annual interest rate
33 U = 500                                           # Domes units per month
34
35 I, K = f.shape
36 hc = h / 365 * p0 * tau                           # in-transit holding $/unit
37
38 # True  -> Domes produces exactly U units ("commission 500 units every month")
39 # False -> Domes produces up to U units
40 DOMES_EXACT = False
41
42 RESULTS = Path("./results")
43 RESULTS.mkdir(exist_ok=True)
44
45
46 # ------------------------------------------------------------------------
47 # Model
```

```python
# ---------------------------------------------------------------------------
def build(name, road=False, domes=False, exact=DOMES_EXACT):
    """Single formulation; Q1/Q2/Q3 just switch road and Domes on or off."""
    m = gp.Model(name)
    m.Params.OutputFlag = 0
    m.Params.MIPGap = 0          # default 1e-4 (~$216 here) can hide the answer

    x = m.addMVar((I, K), vtype=gp.GRB.INTEGER, name="x")   # full containers
    y = m.addMVar(I, vtype=gp.GRB.INTEGER, name="y")        # LCL units
    z = m.addMVar((I, I), vtype=gp.GRB.INTEGER, name="z")   # road i -> j
    v = m.addMVar(I, name="v")                              # Domes units
    w = m.addMVar(I, vtype=gp.GRB.BINARY, name="w")         # Domes site

    ovis = x @ q + y                                        # units landed

    purchase = p0 * ovis.sum()
    freight  = (f * x).sum() + g @ y
    holding  = hc @ ovis
    roadcost = (c * z).sum()
    domescost = p @ v

    m.setObjective(purchase + freight + holding + roadcost + domescost,
                   gp.GRB.MINIMIZE)

    # supply + road in - road out >= demand
    m.addConstr(ovis + v + z.sum(axis=0) - z.sum(axis=1) >= d, name="flow")

    # no shipping a city to itself
    m.addConstr(z[np.arange(I), np.arange(I)] == 0, name="no_self")

    if not road:
        m.addConstr(z == 0, name="no_road")

    if domes:
        m.addConstr(w.sum() == 1, name="one_site")
        if exact:
            m.addConstr(v == U * w, name="domes_cap")
        else:
            m.addConstr(v <= U * w, name="domes_cap")
    else:
        m.addConstr(v == 0, name="no_domes")
        m.addConstr(w == 0, name="no_site")

    m.optimize()
    if m.Status != gp.GRB.OPTIMAL:
        raise RuntimeError(f"{name}: status {m.Status}")

    sol = {k: np.round(var.X) if k != "v" else var.X
           for k, var in dict(x=x, y=y, z=z, v=v).items()}
    return m, sol

```

Question 1 — Current practice: ship everything directly from Ovis (reference)

City	Demand	40' containers	20' containers	LCL units	40' containers ($)	20' containers ($)	LCL units ($)	Total Freight ($)	Purchase (units x $440)	Holding cost ($)	Total cost ($)
Adelaide	420	2	0	20	$4,000.00	$0.00	$500.00	$4,500.00	$184,800.00	$2,278.36	$191,578.36
Brisbane	870	4	0	70	$8,400.00	$0.00	$1,400.00	$7,800.00	$382,800.00	$3,303.82	$393,903.82
Melbourne	1,250	6	0	50	$10,800.00	$0.00	$1,150.00	$11,950.00	$550,000.00	$6,328.77	$568,278.77
Perth	930	4	1	30	$4,800.00	$700.00	$450.00	$5,950.00	$409,200.00	$3,026.96	$418,176.96
Sydney	1,310	6	1	10	$9,900.00	$1,050.00	$220.00	$11,170.00	$576,400.00	$5,921.92	$593,491.92
TOTAL	**4,780**	**22**	**2**	**180**	**$35,900.00**	**$1,750.00**	**$3,720.00**	**$41,370.00**	**$2,103,200.00**	**$20,859.62**	**$2,165,429.62**

Question 2 — Decision variables: sea shipments from Ovis (yellow cells = changing cells for Solver)

City	Demand	40' containers	20' containers	LCL units	40' containers ($)	20' containers ($)	LCL units ($)	Total Freight ($)	Purchase (units shipped x $440)	Holding cost ($)	Total sea cost ($)	Units shipped by sea
Adelaide	420	2	0	20	$4,000.00	$0.00	$500.00	$4,500.00	$184,800.00	$2,278.36	$191,578.36	420
Brisbane	870	4	1	0	$6,400.00	$1,000.00	$0.00	$7,400.00	$396,000.00	$3,417.53	$406,817.53	900
Melbourne	1,250	6	0	30	$10,800.00	$0.00	$690.00	$11,490.00	$541,200.00	$6,227.51	$558,917.51	1,230
Perth	930	4	1	30	$4,800.00	$700.00	$450.00	$5,950.00	$409,200.00	$3,026.96	$418,176.96	930
Sydney	1,310	6	1	0	$9,900.00	$1,050.00	$0.00	$10,950.00	$572,000.00	$5,876.71	$588,826.71	1,300
TOTAL	**4,780**	**22**	**3**	**80**	**$35,900.00**	**$2,750.00**	**$1,640.00**	**$40,290.00**	**$2,103,200.00**	**$20,827.07**	**$2,164,317.07**	**4,780**

Decision variables: units moved by road (row = origin, column = destination)

Origin \ Destination	Adelaide	Brisbane	Melbourne	Perth	Sydney	Total shipped out
Adelaide	0	0	0	0	0	0
Brisbane	0	0	20	0	10	30
Melbourne	0	0	0	0	0	0
Perth	0	0	0	0	0	0
Sydney	0	0	0	0	0	0
Total received	**0**	**0**	**20**	**0**	**10**	
Road cost ($)	**$650.00**					

Demand check: Available = sea + received by road − shipped out by road (must equal Demand)

City	Arrives by sea	Received by road	Shipped out by road	Available	Demand	OK?
Adelaide	420	0	0	420	420	**OK**
Brisbane	900	0	30	870	870	**OK**
Melbourne	1,230	20	0	1,250	1,250	**OK**
Perth	930	0	0	930	930	**OK**
Sydney	1,300	10	0	1,310	1,310	**OK**

Total monthly cost — Question 2

Sea freight ($)	$40,290.00
Purchase from Ovis ($)	$2,103,200.00
Holding cost in transit ($)	$20,827.07
Road transport ($)	$650.00
TOTAL COST — Question 2 ($)	**$2,164,967.07**
Total cost — Question 1 (current practice) ($)	$2,165,429.62
SAVINGS of the best plan vs. current practice ($/month)	**$462.55**
Is the current practice the best?	**NO**